

Phoenix Capsule: Inflatable Aerodynamic Decelerator for re-entry from LEO

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Phoenix 1 Flight

Phoenix 1 was launched in April 2025 on a Falcon 9. After the second stage performed the de-orbit burn, the capsule correctly separated from the stage and established connection with Iridium constellation and ground stations.



Figure 1. Phoenix 1 attached to Falcon 9.

It was a 250 kg satellite with four commercial payloads on board, offering microgravity for biological experiments and in-orbit manufacturing. The mission was meant to demonstrate Atmos knowledge of satellite development and production, as well as developing an Inflatable Aerodynamic Decelerator (IAD).

Phoenix 2 Capsule

Phoenix 2 is a 500 kg satellite that will enhance commercial capabilities of the capsule and bring the IAD development forward

- **100 kg payload** capacity
- **3 hours up to 12 months** flight duration
- **Free flier vehicle** with autonomous propulsion system and de-orbit capability

Atmos’ IAD and EDL System

- **Laminated, cut and stitched in house** The current IAD is fabricated with a laminate fabric of ceramic, metallic and polymeric layers. It acts both as a medium-high temperature TPS and as gas membrane.
- **Rigid Thermal Protection System** High heat flux rate resistant nose, that bears the thermal loads of the stagnation point.
- **Hybrid Inflation System** The volume is initially inflated with stored nitrogen just before the entry interface point. In low subsonic flight, two compressors pump atmospheric air until safe splashdown.
- **Targeted splashdown** The capsule is licensed to splashdown in Portuguese water near Santa Maria Island, Azores. It is equipped with ADS-B to coordinate with local airtraffic.
- **Lifting Body** Lift is achieved with a center of gravity offset with respect to the IAD frame, that trims the vehicle to a designated Angle of Attack. This allows banking turns to steer the vehicle to the designated splashdown area.

Inflatable Heatshield For Mass Saving

Inflatable Heatshields decrease the ballistic coefficient with little mass cost while being stowed in a very small volume. Phoenix 1 had a greater decelerating surface as Apollo capsules and could be stored in a Falcon 9 Rideshare slot. Other projects featuring the same technology are:

1. **LOFTID**, Low-Earth Orbit Flight Test of an Inflatable Decelerator, launched by Nasa in 2022 [1]
2. **EFESTO**, aimed to raise TRL of European Inflatable Heatshield technology up to 5 [1]
3. **ICARUS**, a European project to continue on the path of EFESTO and that aims to bring the technology to TRL 6. A suborbital flight is scheduled in 2028 [3].

Phoenix EDL Technology Development

In the Phoenix development programme, commercial flights are used to rapidly de-risk the subsystems, advancing them directly from TRL 5/6 to 8/9 during operations. While the satellite structure undergoes standard qualification, EDL components follow specialized testing tailored for atmospheric re-entry.

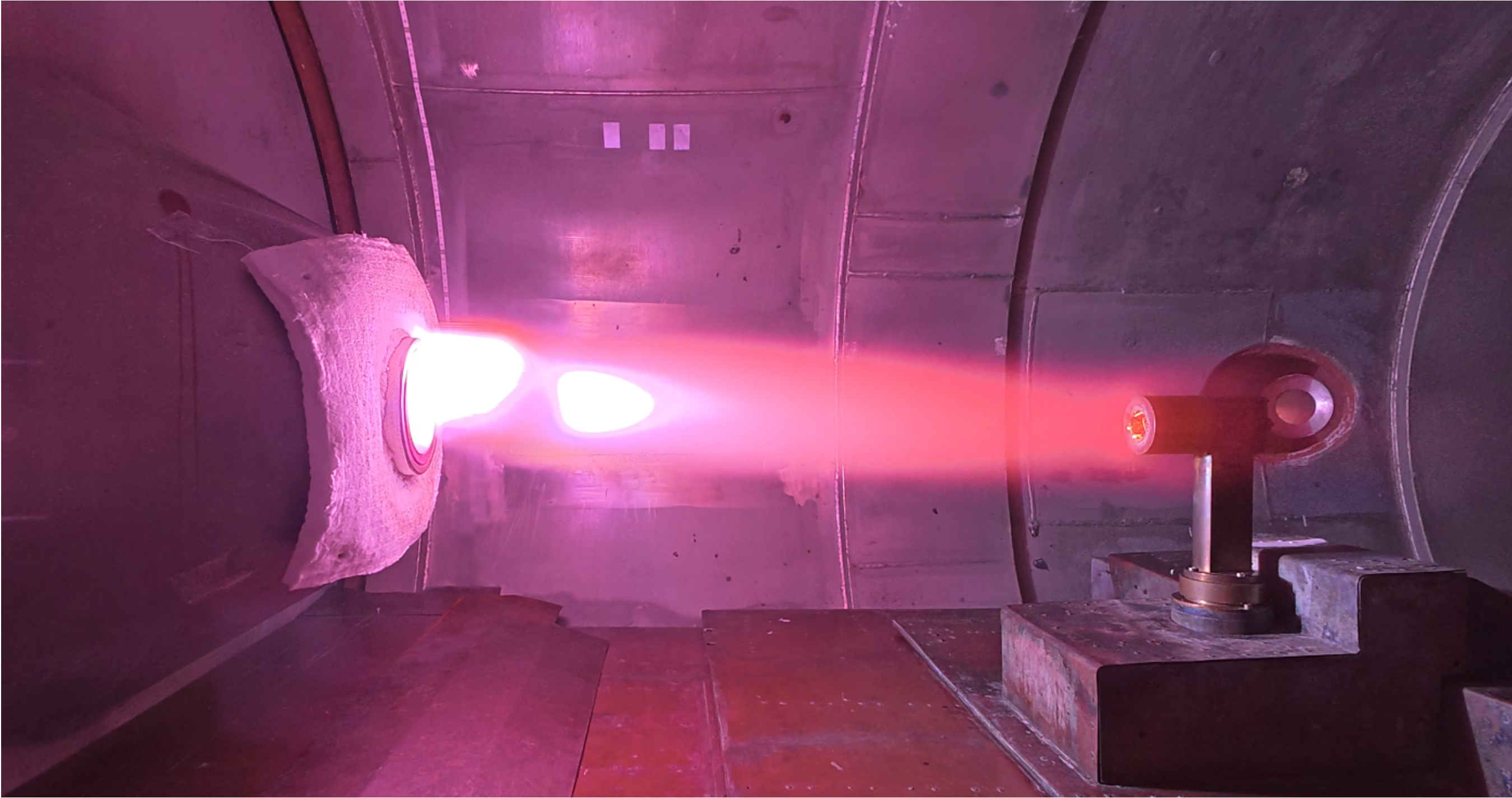


Figure 2. Plasma Wind Tunnel Campaign at IRS in Stuttgart for material qualification.

These testing include and are not limited to:

- **Plasma Wind Tunnel Test** For material degradation tests and qualification of the survivability of critical components after the heat load;
- **Inflation and Static Load Test** To qualify the inflatable structure to the internal and external pressure loads, ensuring against ruptures, excessive leakage and buckling

Fluid Structure Interaction: the Soft Spot of Inflatable Heatshields

Non-rigid structures deform under aerodynamic loads, which, in turn, depend on the aeroshape. Its deformation from nominal can cause deviations on expected heat flux rate and increase uncertainty in trajectories.

The game of forces between external flow and structure can be simulated within the Fluid Structure Interaction framework, in which fluid dynamics and structural equations are iteratively solved.

This implies the solution of equation modeling the non linear behaviour of fabrics, along with Navier-Stokes equations for the external flow. The fabric response must be characterized experimentally.

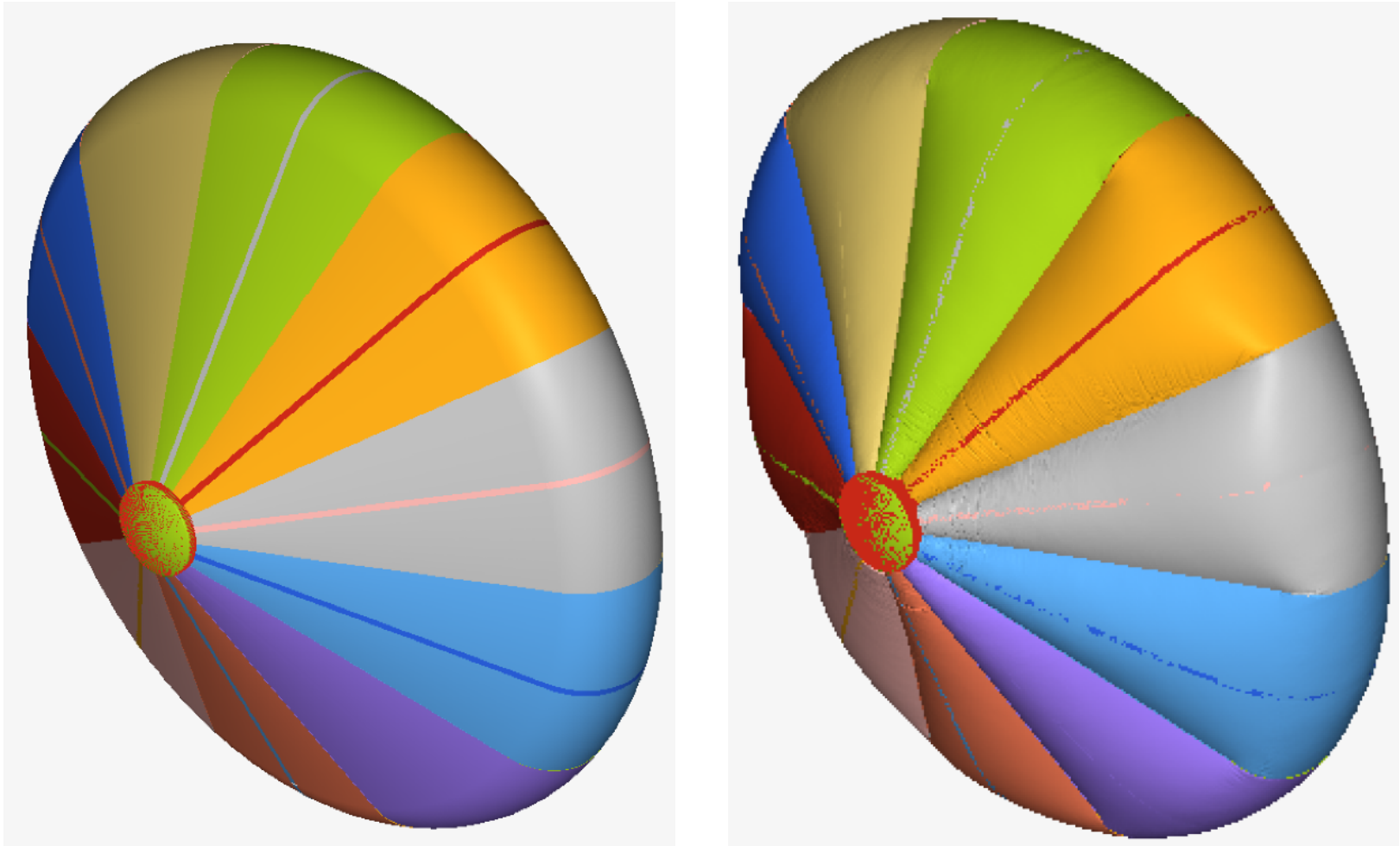


Figure 3. Nominal shape (left) and deformed by internal pressure (right) for the Inflatable Heatshield of ICARUS [4].

Flight Mechanics of the Atmospheric Re-entry of a Lifting Capsule

For a lifting body entering the atmosphere on a shallow trajectory, analytical equations of planar motion can be used for an estimation of peak heat flux rate in the stagnation point and the load factor [5].

$$\dot{q}_{sp,max} \sim \frac{1}{\sqrt{R_N}} \sqrt{\frac{m g}{S c_L}} = \frac{1}{\sqrt{R_N}} \sqrt{g \frac{\beta}{c_L/c_D}} \quad n_{max} \approx \frac{0.905}{c_L/c_D}$$

Where m is the mass, S is the aerodynamic reference surface and g is the gravitational acceleration; R_N is the nose radius, $\beta = \frac{m}{S c_L}$ is the ballistic coefficient and the ratio c_L/c_D is commonly called glide ratio. Chapman model and exponential atmosphere have been included.

Benefits of IADs and of Lift

It is evident that, lowering the ballistic coefficient and increasing the lift coefficient, both the peak stagnation heat flux rate and the load factor will be reduced. Inflatable Aerodynamic Decelerators exploit the reduction of the ballistic coefficient by increasing the drag area. Lifting vehicles allow banking turn maneuvers, to actively control the vehicle trajectory. Nevertheless, gliding trajectories present greater total absorbed heat loads, hence higher temperatures on the backshell of the vehicle (requiring heavier thermal protection systems). Also, if the membrane is leaking, more gas must be stored to sustain a longer flight with a prescribed internal pressure.

Comparison with other classes of entry vehicles

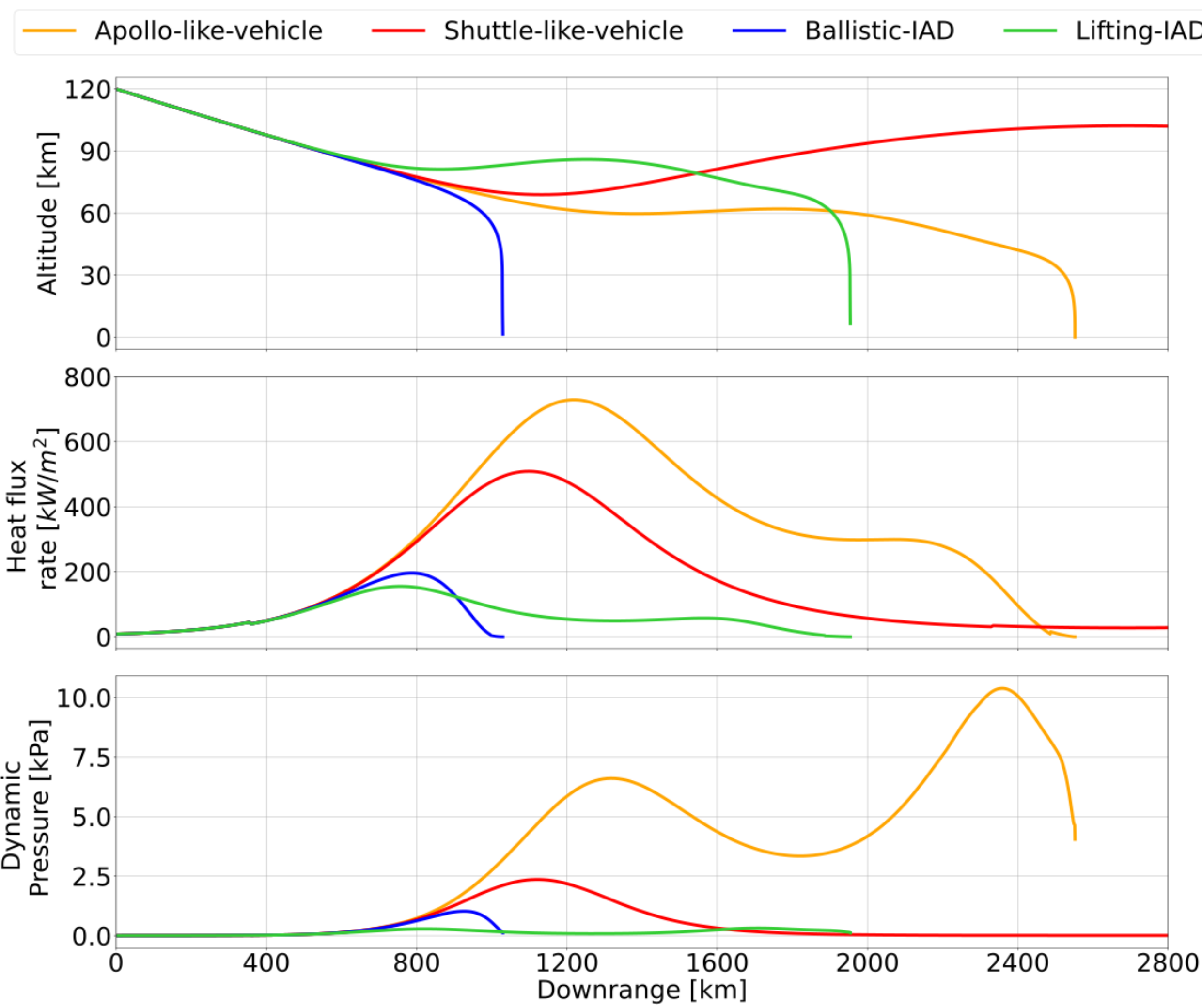


Figure 4. Comparison of performances of different vehicles with identical initial conditions but different β and c_L/c_D .

	Vehicle	Shuttle	B-IAD	L-IAD
β	400	400	12	12
c_L/c_D	0.3	1	0	0.3

References

- [1] DiNonno J. et al., Low-Earth Orbit Flight Test of an Inflatable Decelerator (LOFTID) Mission Overview and Science Return, AIAA Science and Technology Forum and Exposition (2024), Orlando, Florida, USA
- [2] Guidotti G. et al., The EFESTO Project: Advanced European Re-Entry System Based on Inflatable Heat Shield, 2nd International Conference on Flight Vehicles, Aerothermodynamics and Re-Entry Missions (2022), Heilbronn, Germany
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- [4] Banchereau et al., Fluid Structure Interaction Analysis for ICARUS Inflatable Structure, 27th AIAA International Space Planes and Hypersonic Systems and Technologies Conference (2026), Naples, Italy
- [5] Mooij E., Re-entry Systems, Springer, 2024